Field visit kit: Bd de l'Hôpital × Bd Saint-Marcel (Paris 13e)

Model TLS C · Tue 3, Wed 4 or Thu 5 Nov 2026, 07:30-09:30 · print single-sided, bring 2 pens and a clipboard

WHY: the signal program, lane use and turn shares of the model are ASSUMED. This visit measures them and gives the independent checks of spec 13.4 (cycle, queues, turning counts). Never write an estimate: leave a box empty.

Before you go

[ ] Check it is a school-term Tue/Wed/Thu, no strike or roadworks announced (Bd de l'Hôpital, RATP 24/57/91)

[ ] 2 phones (charged, 30+ GB free) on small tripods or clamps, power bank, tally counter or counting app

[ ] This kit printed; a watch or phone showing seconds; a measuring wheel or 10 m tape for distances (optional)

On site, in this order

07:10 Arrive. Set up camera A and B (site map, page 2). Film your phone clock for 5 s at the start of each video: every time on these sheets is PHONE time; write each camera's offset in seconds on the session box below.

07:20 Lane use and crossings questionnaire (page 10), while traffic builds up.

07:30 Start the 15-min counts (pages 7, 8) and keep them going to 09:30 (or count later from the video).

08:00 Signal timing: at least 5 consecutive cycles, ideally 10 (pages 3-6). Reference head: N_straight.

08:00 Bus arrivals and departures at Saint-Marcel - La Pitié (page 11), until 09:00.

During Queues: 2-3 times per approach N, S, W, at the END of red (page 9). Not before 07:35.

09:30 Stop. Photograph every sheet. Copy the videos to the computer the same day.

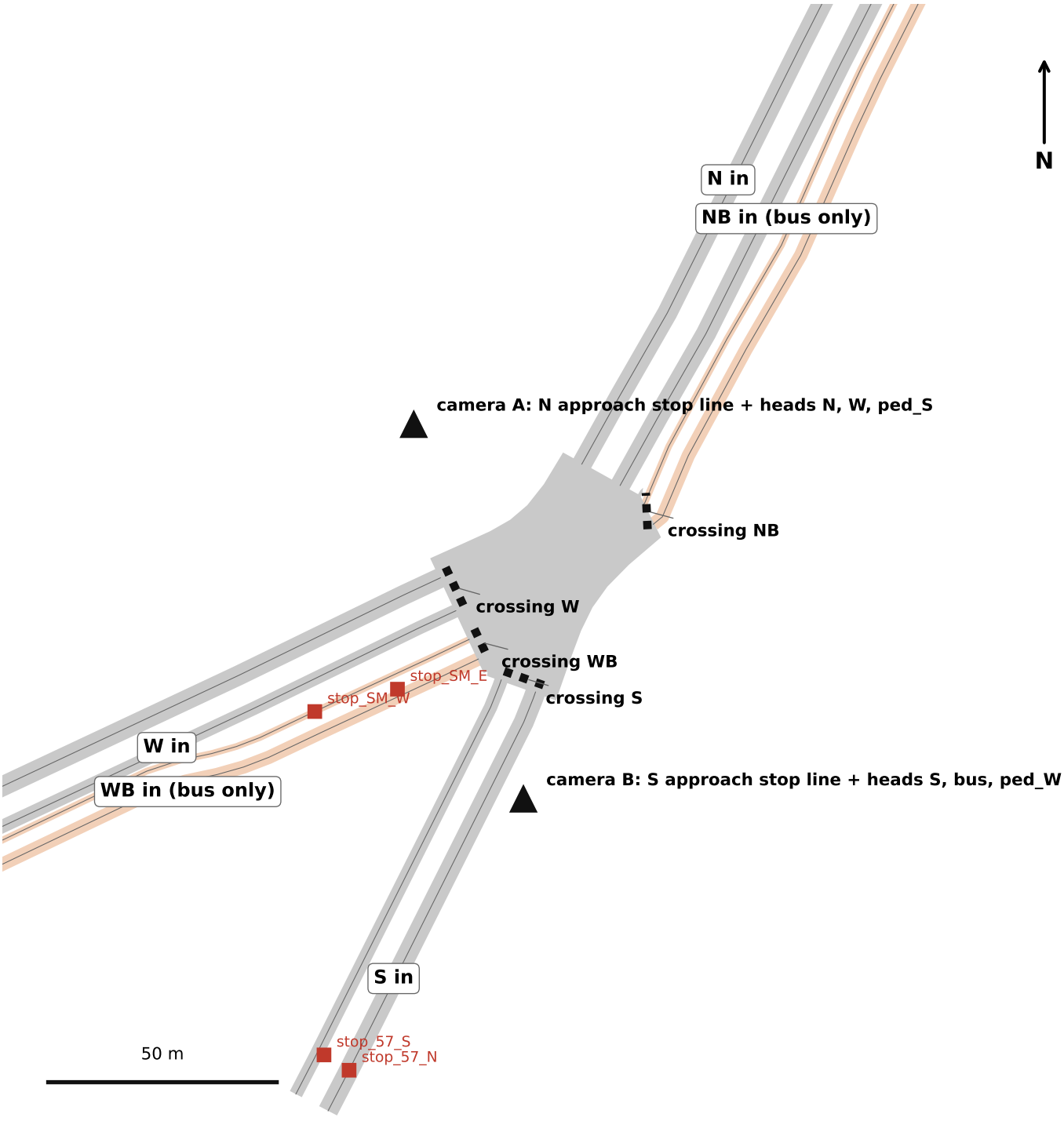
Session

Field	Value
date (YYYY-MM-DD)	
observer(s)	
weather	
camera A position / video file	
camera B position / video file	
camera offset s (camera - phone)	
incidents, roadworks, notes	

AFTER THE VISIT (docs/field_protocol.md): fill data/field/pitie_measurements.yaml, run `tasks.py field-check` until clean, re-download the sensors within a week (`tasks.py data calib-data`), then `tasks.py part3 part4`.
Clock times are quoted strings in the YAML: "08:15:20".

Site map (from the model network)

Approaches N, S, W carry general traffic; NB and WB are bus-only roads. Crossings as in the model; camera spots are suggestions: check both on site.



Model stage order (ASSUMED, check on site): HOP → S_LEFT → STM → BUS_91 → BUS_57; greens (s): HOP 36, S_LEFT 6, STM 20, BUS_91 6, BUS_57 6, plus clearances. Signal heads: N_right, N_straight, bus_91_W, bus_57_S, bus_57_N, S_straight, S_left, bus_91_E, W, ped_NB, ped_S, ped_WB, ped_W.

Signal timing · cycles 1-3

Times to the second (HH:MM:SS, phone time). A cycle starts at the green of N_straight. Pedestrian heads: 'yellow' = start of flashing. A head green twice in a cycle: two lines.

Cycle 1 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Cycle 2 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Cycle 3 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Signal timing · cycles 4-6

Times to the second (HH:MM:SS, phone time). A cycle starts at the green of N_straight. Pedestrian heads: 'yellow' = start of flashing. A head green twice in a cycle: two lines.

Cycle 4 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Cycle 5 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Cycle 6 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Signal timing · cycles 7-9

Times to the second (HH:MM:SS, phone time). A cycle starts at the green of N_straight. Pedestrian heads: 'yellow' = start of flashing. A head green twice in a cycle: two lines.

Cycle 7 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Cycle 8 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Cycle 9 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Signal timing · cycles 10-10

Times to the second (HH:MM:SS, phone time). A cycle starts at the green of N_straight. Pedestrian heads: 'yellow' = start of flashing. A head green twice in a cycle: two lines.

Cycle 10 stage order seen: HOP ___ S_LEFT ___ STM ___ BUS_91 ___ BUS_57 ___ other: _____

Head	Green start	Yellow / flash start	Red start	Note (2nd green, skipped...)
N_straight				
N_right				
bus_91_W				
bus_57_S				
bus_57_N				
S_straight				
S_left				
bus_91_E				
W				
ped_NB				
ped_S				
ped_WB				
ped_W				

Turning counts, 15-min bins (vehicles, cars + vans + trucks)

Count each vehicle once as it leaves the junction. Buses and two-wheelers in their own columns, not in the movements. Tally marks are fine.

Bin	N→S	N→W	S→N	S→W	W→N	W→S	91_E	91_W	24_E	24_W	57_N	57_S	2-wheel
07:30-07:45													
07:45-08:00													
08:00-08:15													
08:15-08:30													
08:30-08:45													
08:45-09:00													
09:00-09:15													
09:15-09:30													

Pedestrians and context flows, 15-min bins

Pedestrians: people starting to cross, both directions together (SHOULD: at least 2 bins). Poliveau: vehicles entering from Rue Poliveau; Wallons: right turns into Rue des Wallons.

Bin	ped NB	ped S	ped WB	ped W	Poliveau in	Wallons right	Note
07:30-07:45							
07:45-08:00							
08:00-08:15							
08:15-08:30							
08:30-08:45							
08:45-09:00							
09:00-09:15							
09:15-09:30							

Queues at the end of red (cycle_max)

2-3 instants per approach, 07:35-09:30. Count queued vehicles at the moment the light turns green; note how far you can see (m) and which lanes.

Approach	Time HH:MM:SS	Vehicles	Visible m	Lanes	Note
N				all	
N				all	
N				all	
S				all	
S				all	
S				all	
W				all	
W				all	
W				all	

Saturation headways (NICE): N straight lane, 3-5 cycles, time each vehicle crossing the stop line from the 4th after g

Approach / lane	Cycle	Crossing times from vehicle 4 (s)
N / lane 2	1	
N / lane 2	2	
N / lane 2	3	
N / lane 2	4	
N / lane 2	5	

Lane use, crossings, loops, ambulance

Answers that change the network. Tick or write; leave blank if unsure.

Question	Answer
Lanes at the N stop line, with arrows (e.g. 2: right | straight)	
Lanes at the S stop line, with arrows	
Lanes at the W (Saint-Marcel) stop line, with arrows	
S approach: length of the bus lane before the stop line (m)	
S approach: do cars use the right lane to go straight? yes / no	
S → W left turn allowed? yes / no protected (own arrow)? yes / no	
Crossing over Bd de l'Hôpital N at the junction itself? yes / no	
Pedestrian-only stage (all cars red, peds green)? yes / no	
Bus stages skipped when no bus? yes / no / unclear	
Greens change from cycle to cycle (actuated)? yes / no	

Crossing	Signalised? (y/n)	Distance to junction m	Length m	Refuge island? (y/n)
NB				
S				
WB				
W				

Loop (saw cut) approach	Lanes	Distance to stop line m	Note

Ambulance	Answer
Route seen (from which arm to which arm)	
Hospital gate used (main gate 83 bd de l'Hôpital? other?)	

Buses at Saint-Marcel - La Pitié (dwell), 08:00-09:00

One line per bus. Line ids as in the model: 91_E, 91_W, 24_E, 24_W, 57_N, 57_S. Arrive = doors open, depart = bus moves off.

Line	Direction (E/W/N/S)	Arrive HH:MM:SS	Depart HH:MM:SS	Note

After the visit: sheet → data/field/pitie_measurements.yaml

Keep the sheet photos with the videos. Run `tasks.py field-check` after each section you fill.

Sheet	YAML section	How
Session box (p. 1)	sessions[0]	date, observer, positions, video, camera_offset_s
Timing (p. 3-6)	signal	cycle_s = gaps between reference green starts
		per head: green_s = yellow - green; yellow_s = red - yellow
		all_red_after_s = red to the next vehicle green
		ped_NB green twice: green_s {HOP: [...], STM: [...]}
		stage_order, varies_between_cycles, bus_stages_on_demand, ped_only_stage
Counts (p. 7)	counts[].movements, buses	one item per 15-min bin: begin, end
Peds (p. 8)	counts[].pedestrians	S, W, WB, NB; poliveau_in, wallons_right
Queues (p. 9)	queues	time, approach, vehicles, visible_m, lanes
Headways (p. 9)	saturation	headways_s list
Lane use (p. 10)	geometry, loops, ambulance	lanes_at_stop_line, crossings.*
Buses (p. 11)	bus_stops	line, direction, arrive, depart